

- 1 A quadratic curve has its minimum point at $(1, -2)$ and has gradient 5 at the point where $x = 2$.
Find the equation of the curve. [5]

Solutions	Comments
Let the equation of the curve be $y = ax^2 + bx + c$. When $x = 1, y = -2$: $a + b + c = -2$ -----(1) $\frac{dy}{dx} = 2ax + b$ When $x = 1, \frac{dy}{dx} = 0$: $2a + b = 0$ -----(2) When $x = 2, \frac{dy}{dx} = 5$: $4a + b = 5$ -----(3) Solving (1), (2) and (3) simultaneously using GC, $a = \frac{5}{2}, b = -5, c = \frac{1}{2}$. Hence, the equation of the curve is $y = \frac{5}{2}x^2 - 5x + \frac{1}{2}$.	

- 2 (a) A sequence is such that $u_1 = p$, where p is a constant, and $u_{n+1} = 2u_n - 5$, for $n > 0$.

(i) Describe how the sequence behaves when

(A) $p = 7$,

[1]

(B) $p = 5$.

[1]

Solutions	Comments
(A) When $p = 7, u_1 = 7, u_2 = 9, u_3 = 13, u_4 = 21, u_5 = 37, \dots$ The sequence is an <u>increasing</u> sequence and it <u>diverges</u>. (B) When $p = 5, u_1 = 5, u_2 = 5, u_3 = 5, \dots$ The sequence is a <u>constant</u> sequence with $u_n = 5$ for all $n > 0$.	'Behaviour of a sequence' – whether the sequence is increasing or decreasing (or neither); converging or diverging.

Note:

Compute the first few terms of the sequence to observe how the sequence behaves.

See screenshots on how to switch to sequence mode:

1. Press MODE and switch to sequence mode.
2. Press Y= to key in sequences. (Press 2ND 7 to key in U.)
3. Press 2ND Graph to see the table of values.

n	u
1	7
2	9
3	13
4	21
5	37
6	69
7	133

(ii) Find the value of p for which $u_5 = 101$.

[2]

Solutions	Comments
$101 = u_5 = 2u_4 - 5 \Rightarrow u_4 = \frac{101+5}{2} = 53$ $53 = u_4 = 2u_3 - 5 \Rightarrow u_3 = \frac{53+5}{2} = 29$ $29 = u_3 = 2u_2 - 5 \Rightarrow u_2 = \frac{29+5}{2} = 17$ $17 = u_2 = 2u_1 - 5 \Rightarrow p = u_1 = \frac{17+5}{2} = 11$	p is u_1 , 101 is u_5 , so use u_5 to find u_4 , then u_3 , ... and finally u_1 .

(b) Another sequence is defined by $v_1 = a$, $v_2 = b$, where a and b are constants, and

$$v_{n+2} = v_n + 2v_{n+1} - 7, \text{ for } n > 0.$$

For this sequence, $v_4 = 2v_3$.

(ii) Find the value of b .

[3]

Solutions	Comments
$v_4 = v_2 + 2v_3 - 7$ $= v_2 + v_4 - 7 \quad (\because v_4 = 2v_3)$ $v_2 = 7$ $\therefore b = v_2 = 7$	Use the definition to work out values of v_1 , v_2 , v_3 , v_4 and v_5 .

(ii) Find an expression in terms of a for v_3 .

[1]

Solutions	Comments
$v_3 = v_1 + 2v_2 - 7 = a + 2(7) - 7 = a + 7$ $v_4 = 2v_3 = 2a + 14$ $v_5 = v_3 + 2v_4 - 7$ $= a + 7 + 2(2a + 14) - 7$ $= 5a + 28$	

(c) The sum of the first n terms of a series is $n^3 - 11n^2 + 4n$, where n is a positive integer.

(i) Find an expression for the n th term of this series, giving your answer in its simplest form.

[2]

Solutions	Comments
$u_n = S_n - S_{n-1}$ $= n^3 - 11n^2 + 4n - ((n-1)^3 - 11(n-1)^2 + 4(n-1))$ $= n^3 - 11n^2 + 4n - ((n^3 - 3n^2 + 3n - 1) - 11(n^2 - 2n + 1) + 4n - 4)$ $= n^3 - 11n^2 + 4n - n^3 + 3n^2 - 3n + 1 + 11n^2 - 22n + 11 - 4n + 4$ $= 3n^2 - 25n + 16$	$u_n = S_n - S_{n-1}$ works for all types of series.

(ii) The sum of the first m terms of this series, where $m > 3$, is equal to the sum of the first three terms of this series. Find the value of m .

[2]

Solutions	Comments
$S_m = S_3$ $m^3 - 11m^2 + 4m = 3^3 - 11(3)^2 + 4(3)$ $m^3 - 11m^2 + 4m + 60 = 0$ Using GC, $m = 10$ or $m = 3$ (rej.) or $m = -2$ (rej.) $\therefore m > 3$.	Remember to reject the wrong values of m .

3 The curve C is defined by the parametric equations

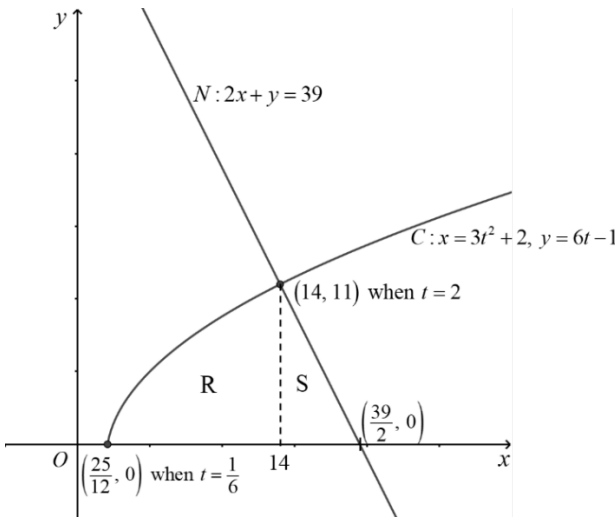
$$x = 3t^2 + 2, \quad y = 6t - 1 \quad \text{where } t \geq \frac{1}{6}.$$

The line N is the normal to C at the point $(14, 11)$.

- (i) Find the cartesian equation of N . Give your answer in the form $ax + by = c$, where a , b and c are integers to be determined. [5]

Solutions	Comments
$\frac{dy}{dt} = 6 \quad \frac{dx}{dt} = 6t$ $\frac{dy}{dx} = \frac{6}{6t} = \frac{1}{t}$ At $(14, 11)$, $y = 14 = 6t - 1 \Rightarrow t = 2$. Gradient of $N = -\frac{1}{\frac{1}{2}} = -2$ Eqn. of N : $y - 11 = -2(x - 14)$ $2x + y = 39$	

- (ii) Find the area enclosed by C , N and the x -axis. [4]

Solutions	Comments
For N : when $y = 0$, $x = \frac{39}{2}$. For C : when $y = 0$, $t = \frac{1}{6}$, $x = \frac{25}{12}$. 	To visualise what the question is asking for, a simple sketch of the two graphs would be useful. The curve C and line N will intersect at point $(14, 11)$. From the sketch we can see that we need to identify the x -intercepts of the two curve C and N with the x -axis. The area enclosed is divided into two separate regions as they are bounded by the two different curves.

Required area = Area of R + Area of S $= \int_{\frac{25}{12}}^{14} y \, dx + \frac{1}{2} \left(\frac{39}{2} - 14 \right) (11)$ $= \int_{\frac{1}{6}}^2 (6t-1)(6t) \, dt + \frac{121}{4}$ $= \frac{2057}{18}$	
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- (iii) The curve C and the line N are both transformed by a 2-way stretch, scale factor 2 in the x -direction and the scale factor 3 in the y -direction, to form the curve D and the line M .

(a) Find the area enclosed by D , M and the x -axis.

[1]

Solutions	Comments
Under the transformations, the point (x, y) will be mapped to $(2x, 3y)$. $\text{Area} = \frac{2057}{18} \times 2 \times 3$ $= \frac{2057}{3}$	All the points on graphs will have its x-coordinates multiplied by 2, the length of the area is multiplied by 2. All y-coordinates are multiplied by 3, the height of the area enclosed is multiplied by 3. Therefore, the enclosed area is multiplied by 2 times and then 3 times.

(b) Find the cartesian equation of D .

[2]

Solutions	Comments
$C: x = 3t^2 + 2, y = 6t - 1, t \geq \frac{1}{6}$ $D: \frac{x}{2} = 3t^2 + 2, \frac{y}{3} = 6t - 1, t \geq \frac{1}{6}$ To find the cartesian equation of D, we can make t the subject: $t = \frac{1}{6} \left(\frac{y}{3} + 1 \right)$ $\therefore \frac{x}{2} = 3 \left(\frac{1}{6} \left(\frac{y}{3} + 1 \right) \right)^2 + 2$ $6x = \left(\frac{y}{3} + 1 \right)^2 + 24$ $54x = (y + 3)^2 + 216$ Since the end-point $\left(\frac{25}{12}, 0 \right)$ on C is mapped to $\left(\frac{25}{6}, 0 \right)$ on D, the graph of D is defined for $y \geq 0$ the cartesian equation of D is $54x = (y + 3)^2 + 216, y \geq 0.$	Stretch by a scale factor of 2 parallel to the x-axis – replace x with $\frac{x}{2}$. Stretch by a scale factor of 3 parallel to the y-axis – replace y with $\frac{y}{3}$. Since the parametric equation has a specific range of t, we have to define a specific range of x or y that will help us to restrict the range or domain of the graph to the defined region.

4

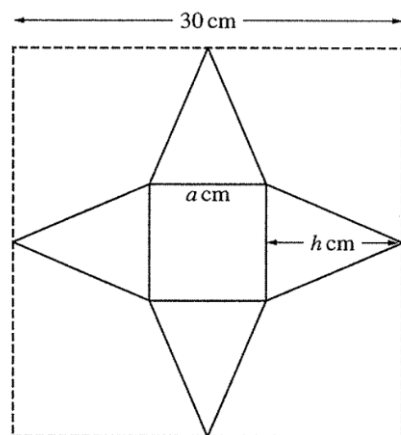


Fig. 1

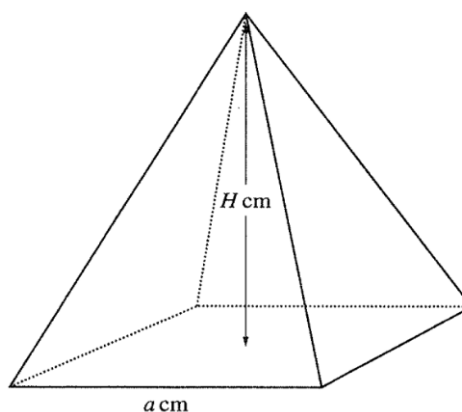


Fig. 2

Fig. 1 shows the net of a square-based pyramid cut from a square of cardboard of side length 30 cm. The net consists of a square of side length a cm and four isosceles triangles, each with base a cm and perpendicular height h cm. The net is folded to form a pyramid which has a square base of side length a cm and vertical height H cm, as shown in Fig. 2.

(i) Show that $H^2 = 225 - 15a$.

[2]

Solutions	Comments
Observe that the slant height of the pyramid is h cm. Applying Pythagoras' theorem, $H^2 = h^2 - \left(\frac{a}{2}\right)^2$ $= \left(\frac{30-a}{2}\right)^2 - \left(\frac{a}{2}\right)^2 \quad \left(\because h = \frac{30-a}{2} \text{ from Fig. 1}\right)$ $= \frac{900 - 60a + a^2 - a^2}{4}$ $= 225 - 15a \quad (\text{shown})$	Can transfer all information useful information onto one diagram to see the relationship between the different variables.

(ii) Find the maximum possible volume of the pyramid. You do not need to show that this value is a maximum. [5]

[The volume of a square-based pyramid is $\frac{1}{3} \times \text{base area} \times \text{height}$.]

Solutions	Comments
$V = \frac{1}{3}a^2\sqrt{225-15a}$ $\frac{dV}{da} = \frac{2}{3}a\sqrt{225-15a} + \frac{1}{3}a^2\left(\frac{-15}{2\sqrt{225-15a}}\right)$ $= \frac{4a(225-15a)-15a^2}{6\sqrt{225-15a}}$ $= \frac{900a-75a^2}{6\sqrt{225-15a}}$ For stationary value of V, $\frac{dV}{da} = 0$. $900a - 75a^2 = 0$ $a(900 - 75a) = 0$ $a = 0 \text{ (rej.) or } a = 12$ Alternative Differentiate both sides wrt to a $V = \frac{1}{3}a^2\sqrt{225-15a} \Rightarrow V^2 = \frac{1}{9}(225a^4 - 15a^5)$ $2V\frac{dV}{da} = \frac{1}{9}(900a^3 - 75a^4) = \frac{1}{9}a^3(900 - 75a)$ For stationary value of V, $\frac{dV}{da} = 0$ so $a = 0 \text{ (rej. since } a > 0) \text{ or } a = 12.$ Maximum volume of pyramid $= \frac{1}{3}(12)^2\sqrt{225-15(12)}$ $= 48\sqrt{45}$ $= 144\sqrt{5} \text{ cm}^3$	Define equation of volume in terms of one variable, a. Note that the question mentioned that 'You do not need to show that this value is a maximum.' There is no need to conduct a first or second derivative test to prove the nature of the stationary point. Same for part 4iiia.

- (iii) (a) Find the value of a for which the total surface area of the four triangular faces of the pyramid is a maximum. You do not need to show that this value is a maximum. [3]

Solutions	Comments
Let $S \text{ cm}^2$ be the total surface area of the four triangular faces. $S = 4 \times \frac{1}{2}ah$ $= 4 \times \frac{1}{2}(a)\left(\frac{30-a}{2}\right)$ $= 30a - a^2$ $\frac{dS}{da} = 30 - 2a$ For stationary value of S, $\frac{dS}{da} = 0$.	

$30 - 2a = 0$ $a = 15$	
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(b) Describe the shape formed from the net in this case. [1]

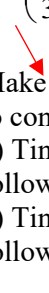
Solutions	Comments
It is a square of side length 15 cm. <i>Note:</i> When $a = 15$, $H = 0$ using part (i).	When the height of the pyramid is 0 cm, the shape formed is a 2-dimensional object. It takes the shape of the base of the pyramid.

- 5 Shania and Tina are playing a game. Shania has a bag containing one green disc, r red discs and $2r$ blue discs, where $r > 1$. A red disc is worth 5 points, a blue disc is worth 2 points and the green disc is worth 0 points. Tina takes two discs from the bag at random. Tina's score is found by multiplying together the number of points for each of the two discs she takes.

(i) State Tina's possible scores. [1]

Solutions	Comments
Tina's possible scores are 0, 4, 10 and 25.	Tina can take 1) 1 G + 1 R: 0 points 2) 1 G + 1 B: 0 points 3) 1 B + 1 B: 4 points 4) 1 R + 1 B: 10 points 5) 1 R + 1 R: 25 points

(ii) Show that the expectation of Tina's scores is $\frac{27r-11}{3r+1}$, and find an expression for the variance. [7]

Solutions					Comments
Let X be the random variable denoting Tina's score.					Let's look at the computation of $P(X=0)$:
Out-come	x	$P(X=x)$	$xP(X=x)$	$x^2P(X=x)$	$P(X=0)$ $= P(1G+1R) + P(1G+1B)$ $= 2\left(\frac{1}{3r+1}\right)\left(\frac{r}{3r}\right) + 2\left(\frac{1}{3r+1}\right)\left(\frac{2r}{3r}\right)$  Make sure that you multiply by 2 to consider the following: 1) Tina draws a green counter followed by a red counter OR 2) Tina draws a red counter followed by a green counter Simplify first before multiplying the probabilities together
GR or GB	0	$2\left(\frac{1}{3r+1}\right)\left(\frac{r}{3r}\right) + 2\left(\frac{1}{3r+1}\right)\left(\frac{2r}{3r}\right)$ $= \frac{6}{3(3r+1)}$	0	0	
BB	4	$\left(\frac{2r}{3r+1}\right)\left(\frac{2r-1}{3r}\right) = \frac{4r-2}{3(3r+1)}$	$\frac{16r-8}{3(3r+1)}$	$\frac{64r-32}{3(3r+1)}$	
RB	10	$2\left(\frac{r}{3r+1}\right)\left(\frac{2r}{3r}\right) = \frac{4r}{3(3r+1)}$	$\frac{40r}{3(3r+1)}$	$\frac{400r}{3(3r+1)}$	
RR	25	$\left(\frac{r}{3r+1}\right)\left(\frac{r-1}{3r}\right) = \frac{r-1}{3(3r+1)}$	$\frac{25r-25}{3(3r+1)}$	$\frac{625r-625}{3(3r+1)}$	
		1	$\frac{81r-33}{3(3r+1)}$	$\frac{1089r-657}{3(3r+1)}$	

$E(X) = \sum_x xP(X=x)$ $= 0 + \frac{16r-8}{3(3r+1)} + \frac{40r}{3(3r+1)} + \frac{25r-25}{3(3r+1)}$ $= \frac{81r-33}{3(3r+1)}$ $= \frac{27r-11}{3r+1} \quad (\text{shown})$ $E(X^2) = \sum_x x^2 P(X=x)$ $= 0 + \frac{64r-32}{3(3r+1)} + \frac{400r}{3(3r+1)} + \frac{625r-625}{3(3r+1)}$ $= \frac{1089r-657}{3(3r+1)}$ $= \frac{363r-219}{3r+1}$ $\text{Var}(X) = E(X^2) - [E(X)]^2$ $= \frac{363r-219}{3r+1} - \left(\frac{27r-11}{3r+1} \right)^2$ $= \frac{(363r-219)(3r+1) - (27r-11)^2}{(3r+1)^2}$ $= \frac{1089r^2 - 294r - 219 - 729r^2 + 594r - 121}{(3r+1)^2}$ $= \frac{360r^2 + 300r - 340}{(3r+1)^2}$	To find the variance, $\text{Var}(X) = E(X^2) - [E(X)]^2$ we need to find $E(X) = \sum_x xP(X=x)$ $E(X^2) = \sum_x x^2 P(X=x)$
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(iii) Given that the variance of Tina's score is 38, find the value of r .

[2]

Solutions	Comments
$\frac{20(18r^2 + 15r - 17)}{(3r+1)^2} = 38$ $10(18r^2 + 15r - 17) = 19(9r^2 + 6r + 1)$ $9r^2 + 36r - 189 = 0$ $r^2 + 4r - 21 = 0$ $(r+7)(r-3) = 0$ $r = -7 \text{ (rej.) or } r = 3$	It is given in the question that $r > 1$; r also cannot be negative.

- 6 In this question you should assume that T , W and D follow independent normal distributions. James leaves home to go to work at T minutes past 8 am each day, where T follows the distribution $N(5, 1.2^2)$.

(i) Sketch this distribution for the period from 8 am to 8.10 am.

[2]

Solutions	Comments
$T \sim N(\mu, \sigma^2), \mu = 5, \sigma = 1.2$ $\mu - 3\sigma = 1.4$ $\mu + 3\sigma = 8.6$ $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$	By the empirical rule of the normal distribution, we would expect approximately 99.7% of the data to lie within 3 standard deviation from the mean. $5 \pm 3(1.2)$ have a range of values from 1.4 to 8.6. Since you are required to draw the distribution from 8am to 8.10am, and 99.7% of the data lies between 1.4 and 8.6, you have to ensure that the area under the distribution is close to 0 from 0 to 1.4 and 8.6 to 10.

- (ii) Find the probability that, on a randomly chosen day, James leaves for work later than 8.06 am. [1]

Solutions	Comments
$P(T > 6) = 0.20233 = 0.202$ (to 3 s.f.)	

When the weather is fine, James walks to work. The time, W minutes, he takes to walk to work follows the distribution $N(21, 3^2)$. James is supposed to start work at 8.30 am.

- (iii) Find the probability that, on a randomly chosen day when James walks, he is late for work. [2]

Solutions	Comments
$E(T + W) = 5 + 21 = 26$ $\text{Var}(T + W) = 1.2^2 + 3^2 = 10.44$ $T + W \sim N(26, 10.44)$ $P(T + W > 30) = 0.10786$ $= 0.108$ (to 3 s.f.)	For James to be late, he needs to reach work after 8.30 am (i.e. $T + W > 30$).

When the weather is not fine, James drives to work. He still leaves at T minutes past 8 am each day; the time, D minutes, he takes to drive to work follows the distribution $N(19, 6^2)$.

On average, the weather is fine on 70% of mornings.

- (iv) One day, James is late for work. Find the probability that the weather is fine that day. [5]

Solutions	Comments
$E(T + D) = 5 + 19 = 24$ $\text{Var}(T + D) = 1.2^2 + 6^2 = 37.44$ $T + D \sim N(24, 37.44)$	We are looking for conditional probability here. We are given the information that James is late for work. So we are finding the probability that the weather is

$P(T + D > 30) = 0.16340$ fine that day given that James is late for work. We need to find the probability that on a randomly chosen day when James drives, he is late for work. Hence, find $P(T + W > 30)$ Draw a tree diagram to help you visualise. Use at least 5sf in the answer from part (iii)	
$P(\text{weather is fine} \mid \text{James is late for work})$ $= \frac{P(\text{weather is fine AND James is late for work})}{P(\text{James is late for work})}$ $= \frac{(0.7)(0.10786)}{(0.7)(0.10786) + (0.3)(0.16340)}$ $= 0.606 \quad (\text{to 3 s.f.})$	

- 7 A study into the germination of parsnip seeds in the 1980s produced the following data for the average number of days, d , taken for a seed to germinate at different soil temperature, t , measured in degrees Fahrenheit.

t	32	41	50	59	68
d	172	57	27	19	14

- (i) Sketch a scatter diagram of the data. State the product moment correlation coefficient between d and t . [2]

Solutions	Comments
Using GC, $r = -0.848$ (to 3 s.f.).	

- (ii) Lim thinks the data can be modelled by the regression equation $d = -a + bu$, where $u = \frac{1}{t}$. Find the values of a and b for Lim's model, giving the values correct to 3 significant figures. State the product moment correlation coefficient between d and u . [3]

Solutions						Comments
t	32	41	50	59	68	
$u = \frac{1}{t}$	$\frac{1}{32}$	$\frac{1}{41}$	$\frac{1}{50}$	$\frac{1}{59}$	$\frac{1}{68}$	
d	172	57	27	19	14	
For the model $d = -a + bu$, we regress d on u. Using GC, the regression line of d on u is given by $d = -145.12547 + 9456.40257u$ and $r = 0.94123$. To 3 s.f., $a = 145$, $b = 9460$ and $r = 0.941$.						

The study also found that, at a soil temperature of 86 degrees Fahrenheit, parsnip seeds took an average of 32 days to germinate.

(iii) Determine whether Lim's model fits this additional data. [1]

Solutions	Comments
According to Lim's model, when $t = 86$, $d = -a + bu = -145 + 9460\left(\frac{1}{86}\right) = -35 \neq 32.$ This is not possible and hence, Lim's model does not fit this additional data. Moreover, since $t = 86$ lies outside the range of data, $32 \leq t \leq 68$, Lim's model is not suitable for extrapolation.	

A temperature of F degrees Fahrenheit is equivalent to a temperature of C degrees Celsius, where

$$C = \frac{5}{9}(F - 32).$$

(iv) Write Lim's equation from part (ii) in terms of d and T , where T is the temperature in degrees Celsius. [2]

Solutions	Comments
$T = \frac{5}{9}(t - 32) \Rightarrow t = \frac{9}{5}T + 32$ $d = -145.12547 + 9456.40257u$ $= -145.12547 + \frac{9456.40257}{t}$ $= -145.12547 + \frac{9456.40257}{\frac{9}{5}T + 32}$ $= -145.12547 + \frac{47282.01285}{9T + 160}$ $\approx -145 + \frac{47300}{9T + 160}$	

- 8 In a game, a computer randomly chooses 12 shapes from 11 circles and 17 rectangles. The number of rectangles chosen is denoted by R .

(i) Show that $P(R=1) < P(R=2)$. [2]

Solutions	Comments
$P(R=1) = \frac{{}^{17}C_1 \times {}^{11}C_{11}}{{}^{28}C_{12}} = 5.59 \times 10^{-7}$ $P(R=2) = \frac{{}^{17}C_2 \times {}^{11}C_{10}}{{}^{28}C_{12}} = 4.92 \times 10^{-5}$ $\therefore P(R=1) < P(R=2)$ Alternative solution $\frac{P(R=1)}{P(R=2)} = \frac{\frac{{}^{17}C_1 \times {}^{11}C_{11}}{{}^{28}C_{12}}}{\frac{{}^{17}C_2 \times {}^{11}C_{10}}{{}^{28}C_{12}}} = \frac{1}{88} < 1$ $\therefore P(R=1) < P(R=2)$	Probability tree method is possible but less efficient as there will be 12 choices.

The number of rectangles available is now increased by r . The computer randomly chooses 12 shapes from the 11 circles and $(17+r)$ rectangles. The probability that 4 rectangles are chosen is now 15 times the probability that 3 rectangles are chosen.

(ii) Find the value of r . [5]

Solutions	Comments
$P(R=4) = 15 \times P(R=3)$ $\frac{{}^{17+r}C_4 \times {}^{11}C_8}{{}^{28+r}C_{12}} = 15 \times \frac{{}^{17+r}C_3 \times {}^{11}C_9}{{}^{28+r}C_{12}}$ ${}^{17+r}C_4 \times {}^{11}C_8 = 15 \times {}^{17+r}C_3 \times {}^{11}C_9$ $\frac{(17+r)!}{4!(13+r)!} \times \frac{11!}{8!3!} = 15 \times \frac{(17+r)!}{3!(14+r)!} \times \frac{11!}{9!2!}$ $\frac{1}{4!(13+r)!8!} = \frac{15}{(14+r)!9!2!}$ $\frac{(14+r)!9!}{(13+r)!8!} = \frac{15 \times 4!}{2!}$ $9(14+r) = 180$ $14+r = 20$ $r = 6$	MF26: ${}^nC_r = \frac{n!}{r!(n-r)!}$

- 9 A factory produces ballpoint pens. On average 6% of the pens are faulty. The pens are packed in boxes of 100 for sale to retail outlets. It should be assumed that the number of faulty pens in a box of 100 pens follows a binomial distribution.

For quality control purposes a random sample of 10 pens from each box is tested. If 2 or fewer faulty pens are found in the sample of 10, the box is accepted for sale. Otherwise the box is rejected.

(i) Explain what is meant by a random sample in this context. [1]

Solutions	Comments
A random sample in this context means that each pen in the box of 100 has an equal probability of being selected, and the selections are independent of each other.	There are 2 ideas that need to be brought across unambiguously.

- (ii) Find the probability that a randomly chosen box of 100 pens is accepted for sale. [1]

Solutions	Comments
Let X be the number of faulty pens in a sample of 10. $X \sim B(10, 0.06)$ $P(X \leq 2) = 0.981162$ $= 0.981 \text{ (to 3 s.f.)}$	

- (iii) One morning 75 boxes are tested in this way. Find the probability that more than 5% of these boxes are rejected. [4]

Solutions	Comments
Let Y be the number of boxes, out of 75, that are rejected. $P(\text{a box is rejected}) = 1 - 0.981162$ $= 0.018838$ $Y \sim B(75, 0.018838)$ $P\left(Y > \frac{5}{100} \times 75\right) = P(Y > 3.75)$ $= 1 - P(Y \leq 3)$ $= 0.0535 \text{ (to 3 s.f.)}$	Define your random variables clearly. We are counting number of pens in (ii) and the number of boxes in (iii)

An alternative testing procedure is trialled in which a random sample of 5 pens is initially taken from a box and tested.

- If there are no faulty pens in this sample of 5 the box is accepted.
- If there are 3 or more faulty pens in this sample of 5 the box is rejected.
- If there are 1 or 2 faulty pens in this sample a second random sample of 5 pens is taken from the box. When the second sample has been tested, the box is accepted if the **total** number of faulty pens found in the combined sample of 10 is 2 or fewer and rejected otherwise.

- (iv) Find the probability that a randomly chosen box of 100 pens is accepted for sale when the alternative testing procedure is used. [5]

Solutions	Comments
Let W_1 and W_2 be the number of faulty pens in the first and second samples of 5 respectively. $W_1 \sim B(5, 0.06)$ and $W_2 \sim B(5, 0.06)$	Draw a probability tree to help visualize the alternative testing procedure. This will help us visualize what are the cases for accepting the box. Keep to at least 5sf for intermediate working to get an accurate final answer.

$W_1 = 0$ (Box accepted) $W_1 = 1$ (2nd sample taken) $W_1 = 2$ (2nd sample taken) $W_1 \geq 3$ (Box rejected) $W_2 = 0 \text{ or } 1$ (Box accepted) $W_2 \geq 2$ (Box rejected) $W_2 = 0$ (Box accepted) $W_2 \geq 1$ (Box rejected) $P(\text{a box is accepted})$ $= P(W_1 = 0) + P(W_1 = 1)P(W_2 \leq 1) + P(W_1 = 2)P(W_2 = 0)$ $= 0.983 \quad (\text{to 3 s.f.})$	
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- (v) Explain why the factory manager might prefer to use the alternative testing procedures. [1]

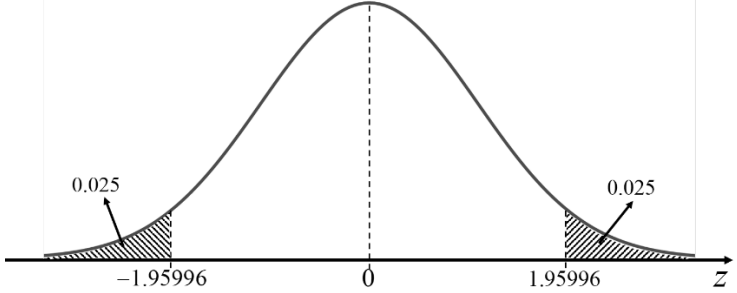
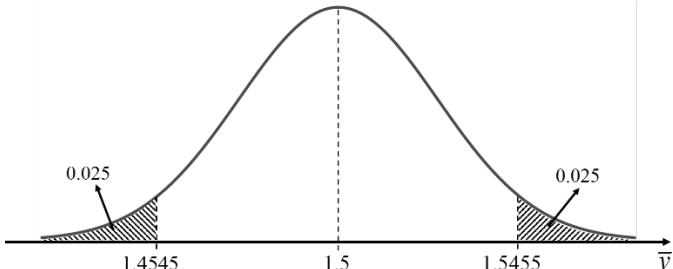
Solutions	Comments
The factory manager might prefer to use the alternative testing procedure since the expected number of pens tested is less than 10. Hence, the alternative testing procedure saves time and cost in the process of testing.	The manager is sometimes able to accept/reject the box by testing only 5 pens ($W_1 = 0$ or $W_1 = 3$), instead of always testing 10 pens in the original test

- 10 Carbon steel is made by adding carbon to iron; this makes the iron stronger, though less flexible. A steel manufacturing company makes carbon steel in the form of round bars. The process is designed to manufacture bars in which the amount of carbon in each bar, by weight, is 1.5%. It is known that the percentage of carbon in the steel bars is distributed normally, and that the standard deviation is 0.09%.

After comments from customers, the production manager wishes to test, at the 5% level of significance, if the percentage of carbon in the steel bars is, in fact, 1.5%. He examines a random sample of 15 bars to determine the percentage of carbon in each bar.

- (i) Find the critical region for this test. [4]

Solutions	Comments
Let $Y\%$ be the percentage of carbon in a round bar. Let $\mu_y\%$ be the population mean percentage of carbon in round bars. $H_0 : \mu_y = 1.5$ $H_1 : \mu_y \neq 1.5$ Level of significance: 5% Test statistic: When H_0 is true, $Z = \frac{\bar{Y} - 1.5}{0.09/\sqrt{n}} \sim N(0,1)$.	Note that the unit here is percentage and there is no need to convert to a number. We use Y as the subsequent question part already uses X. Critical region is also known as the rejection region. Take note that the critical region should be in terms of $\bar{y}$ and not z.

	Recall from sampling $E(\bar{Y}) = E(Y)$ $\text{Var}(\bar{Y}) = \frac{1}{n} \text{Var}(Y)$ Note: To find the critical values using invNorm, remember to key in $\sigma = \frac{0.09}{\sqrt{15}}$
Critical region: $z \leq -1.95996$ or $z \geq 1.95996$ $\frac{\bar{y} - 1.5}{0.09/\sqrt{15}} \leq -1.95996$ or $\frac{\bar{y} - 1.5}{0.09/\sqrt{15}} \geq 1.95996$ $\bar{y} \leq 1.4545$ or $\bar{y} \geq 1.5455$ Therefore, the critical region is $\{\bar{y} \in \mathbb{R} : \bar{y} \leq 1.45 \text{ or } \bar{y} \geq 1.55\}$.	
Alternative solution Let $Y\%$ be the percentage of carbon in a round bar. Let $\mu_y\%$ be the mean percentage of carbon in round bars. $H_0 : \mu_y = 1.5$ $H_1 : \mu_y \neq 1.5$ Level of significance: 5% Test statistic: When H_0 is true, $\bar{Y} \sim N\left(1.5, \frac{0.09^2}{15}\right)$. 	
For H_0 to be rejected, $P(\bar{Y} \leq \bar{y}) \leq 0.025$ or $P(\bar{Y} \geq \bar{y}) \leq 0.025$. Using GC, $\bar{y} \leq 1.4545$ or $\bar{y} \geq 1.5455$. Therefore, the critical region is $\{\bar{y} \in \mathbb{R} : \bar{y} \leq 1.45 \text{ or } \bar{y} \geq 1.55\}$.	

The company recently launched a new line of flat bars made from mild steel. In mild steel the amount of carbon in each bar, by weight, is 0.25%.

Comments from customers suggest that these bars are not sufficiently flexible, which makes the production manager suspect that too much carbon has been added. He decides to perform a hypothesis test on a random sample of 40 of the new flat bars to find out if this is the case.

- (ii) Explain why the production manager takes a sample of 40 flat bars for this test when he only took a sample of 15 round bars in his earlier test. [2]

The percentage of carbon in the flat steel bars may not follow a normal distribution. Taking a sample of 40 is large enough for the manager to apply Central Limit Theorem to **approximate** the **sample mean** percentage of carbon in flat steel bars using a normal distribution.

Furthermore, s^2 , the unbiased estimate of the population variance would be a good estimate for the population variance.

The amounts of carbon, $x\%$, in a random sample of 40 flat bars are summarised as follows.

$$n = 40 \quad \sum x = 10.16 \quad \sum x^2 = 2.586342$$

- (iii) Calculate unbiased estimates of the population mean and variance for the percentage amount of carbon in the flat bars. [2]

Solutions	Comments
Unbiased estimate for population mean, $\bar{x} = \frac{10.16}{40} = 0.254$. Unbiased estimate for population variance, $s^2 = \frac{1}{39} \left(2.586342 - \frac{10.16^2}{40} \right)$ $= 0.00014621$ $= 0.000146 \text{ (to 3 s.f.)}$	

- (iv) Test, at the $2\frac{1}{2}\%$ level of significance, whether the mean amount of carbon in the flat bars is more than 0.25%. You should state your hypotheses and define any symbols that you use. [5]

Solutions	Comments
Let $X\%$ be the percentage of carbon in a flat bar. Let $\mu_x\%$ be the population mean percentage of carbon in flat bars. $H_0 : \mu_x = 0.25$ $H_1 : \mu_x > 0.25$ Level of significance: 2.5% Test statistic: When H_0 is true, since $n = 40$ is large, by Central Limit Theorem, $\bar{X}$ is approximately normally distributed. $Z = \frac{\bar{X} - 0.25}{S / \sqrt{n}} \sim N(0,1) \text{ approximately.}$ Computation: $n = 40$, $\bar{x} = 0.254$, $s^2 = 0.00014621$. Using G.C. $p\text{-value} = 0.018211$. Conclusion: Since $p\text{-value} = 0.0182 < 0.025$, H_0 is rejected at the 2.5% level of significance. There is sufficient evidence to conclude that the population mean percentage of carbon in flat steel bars is more than 0.25% at the 2.5% level of significance.	